AI Application on Idea Generation for Seafood Alternative Products

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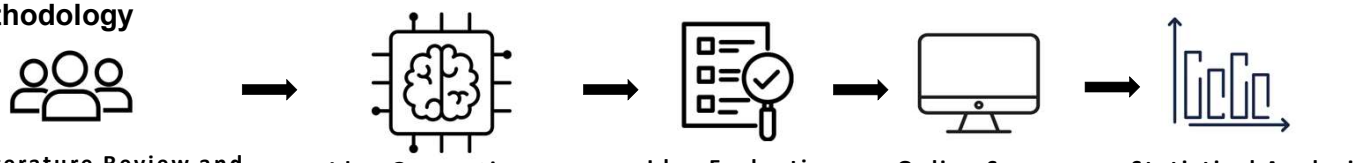
Introduction

- **Rising global seafood demand versus unsustainable practices** (overfishing, climate change impacts)[1; 2; 3]
- **Plant-based seafood as a sustainability solution** (reduced pressure on marine resources, innovation challenges)[4]
- **Increasing use of artificial intelligence in the corporate sector** (e.g. innovative product development)[5]
- **Research gap:** Lack of comparative research on AI-generated versus human-generated ideas in the development of vegan seafood products

Objectives

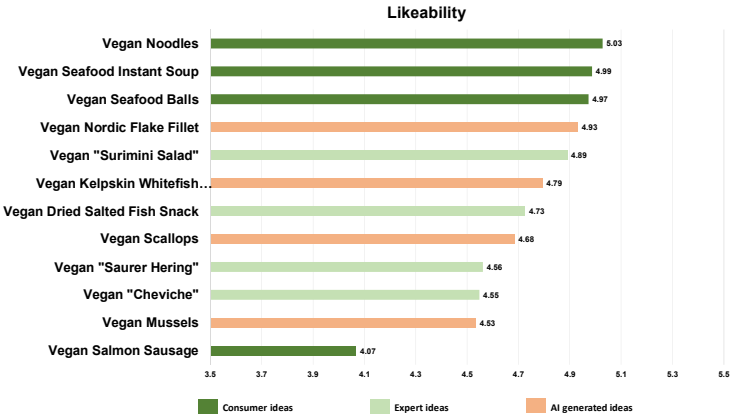
- Evaluate AI's potential for idea generation in food product development process
- Compare AI-driven concepts with human creativity
  - Innovation, likeability and willingness to buy (WTB)

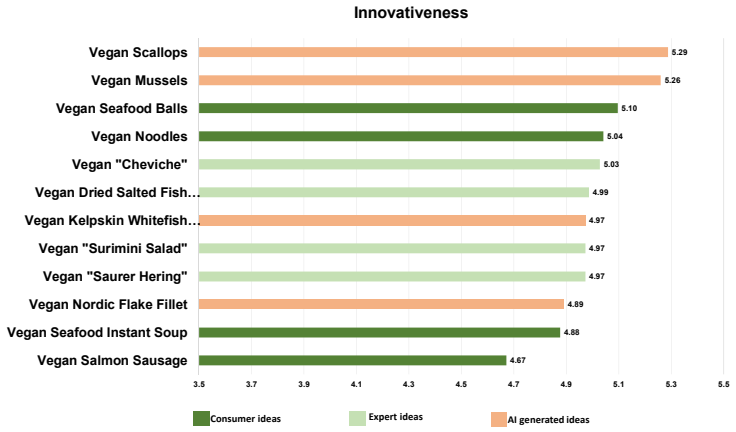
Methodology



Results

Total Participants = 73 → Omnivore = 26 Flexitarian = 23 Vegetarian + Vegan = 20 Others = 4







Parameter	WTB	Likeability
WTB	1	0,9585
Likeability	0,9585	1

Table 1. WTB – Likeability correlation

There is no significant difference among the 3 different groups, based on the results of Anova

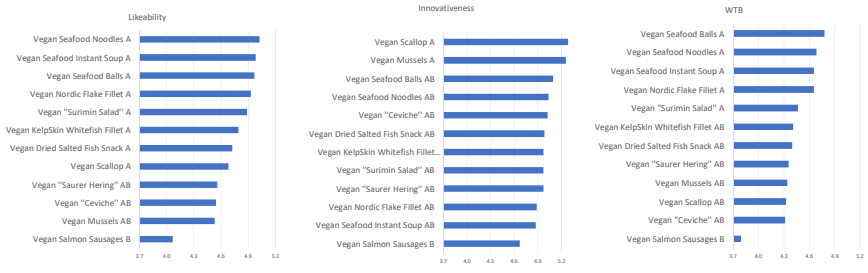
Discussion

- No effect caused by familiarity to vegan products; according to results of T-test
- Based on results of Anova → no significant difference in analysis of different parameters..
- Anova test on diets:

Comparison A vs B	Likeability	WTB	Innovativeness
Flexitarian vs Omnivore	0.245	0.008 (Flex Higher)	0.148
Flexitarian vs Veg + Vegetarian	0.675	0.936	0.983
Omnivore vs Veg + Vegetarian	0.046 (Veg Higher)	0.004 (Veg Higher)	0.118

Table 2. Results of Anova post-hoc Tukey test ; x<0.05 = significance threshold

- Pairwise comparison of ideas (Duncan test):



Conclusion

- The potential of AI; High innovation but lacks consumer awareness.
- The lack of market perspective in AI idea development
- No influence of familiarity on consumer perception in different ideas
- Diets influence the perception of the 12 new vegan seafood products.

Limitations and Outlook

Limitations:

- Low number of participants
- Low number of vegan participants → merge of vegan and vegatarian participants

Outlook:

- AI can be used to level up new ideas with regard to innovativeness
- There can not be still a mere reliance on AI to provide both Innovative and consumer-liked new ideas.

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